

*Directions: Decide whether the situation requires a permutation or combination. Then, answer the question either using the formulas or the counting principle. Show your work either way!*

- a) A television news director wishes to use three news stories on an evening show. One story will be the “lead story”, one will be the second story, and the last will be a “closing story”. If the director has a total of eight stories to choose from, how many possible ways can the program be set up?
- b) A guitar shop owner has 12 guitars in his main shop display. The owner wishes to select 5 of them to display at a special show. How many different ways can a group of 5 guitars be selected?
- c) In a board of directors composed of 10 people, how many ways can a chief executive officer, a director, and a treasurer be selected?
- d) A radio station has 10 songs they need to play in the morning. If they play all the songs, how many different orders can the songs be played in?
- e) You are going on vacation and need to choose some books to take with you. You have 9 books you want to read but only 5 will fit in your suitcase. How many different collections of books can you take with you?
- f) Your school is putting on a production of Beauty and the Beast. The three main roles are Belle, Beast, Gaston, Lumiere, and Mrs. Potts. If 15 people audition for the roles, how many different ways are there to fill the spots?